The fastest particle passes Q at (3 cm,3 cm). By (18)、(19), greatest entry speed V_1' is 3 times least V_2; their circular radii R_1' and R_2 in the field obey

$$R_1' = 3R_2$$

The circle for speed V_1' passes origin O, tangent to x, and passes Q, giving

$$R_1' = 3 \text{ cm}$$

Center O_1' is at (0 cm,3 cm). The slower orbit has radius

$$R_2 = 1 \text{ cm}$$

O_2 lies at (0 cm,1 cm); its circle is

$$x^2 + (y-1)^2 = 1 \tag{20}$$

Draw a tangent from Q, touching P at (x_P, y_P). If it exits the field at P with V_2, it follows PQ to Q. P must lie on the field boundary.

Find P. O_2P is normal to PQ. Drop P onto $O_1'Q$ at H; drop O_2 onto PH at F. Triangles $\triangle O_2PF$ and $\triangle PQH$ are similar, so

$$\frac{\overline{O_2F}}{\overline{PF}} = \frac{\overline{PH}}{\overline{HQ}} \tag{21}$$

i.e.

$$\frac{x_P}{(1-y_P)} = \frac{(3-y_P)}{(3-x_P)} \tag{22}$$

giving $$x_P^2 + y_P^2 - 4y_P = 3x_P - 3$$

Completing the square:

$$x_P^2 + (y_P - 1)^2 - 2y_P - 1 = 3x_P - 3$$

Point P at coordinates (x_P, y_P) satisfies (20),

$$-2y_P = 3x_P - 3$$

$$y_P = \frac{3}{2}(1 - x_P) \tag{23}$$

Using (20), the y coordinate of P is

$$y_P = 0.354 \text{ cm} \tag{24}$$

Thus shift boundary MN down parallel through P to $M'N'$ as shown. At V_2 it exits at P along the circle tangent to reach Q.

6.

As a、b move right, b and c first meet and collide. Impact is brief: both springs retain their natural lengths. a、d are unchanged; b、c have equal mass. Elastic impact leaves b at rest and c at v_0, rightward. Just after impact, a has v_0 rightward; b rests. Mass center M_1 of a and b is the spring midpoint, at $v_0/2$ rightward; c has v_0 rightward, d rests. For c and d, mass center M_2 is also the spring midpoint, at $v_0/2$ rightward.

Origin M_1 follows the mass center rightward at $v_0/2$. Its X' axis joins all four centers (mass-center frame). Take the first b and c impact as the time